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ITMM 485 / 585
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Legal and Ethical Issues in Information Technology

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Ch11: Online Communities, Virtual Reality, And Artificial Intelligence

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Online Communities, Virtual Reality, and Artificial Intelligence

1) *online communities*, including social networking services (SNSs);
2) *virtual environments* (*VEs*), including virtual reality (VR) applications;
3) *artificial intelligence* (*AI*).

- A unifying theme that brings together this disparate cluster of topics is the impact they have for our notions of *community* and (personal) *identity* in the digital age.

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Online Communities, Virtual Reality, and Artificial Intelligence (Continued)

- The two main themes, community and (personal) identity, are analyzed via the three distinct aspects of cybertechnology in the following ways:

a) SNSs enable social interactions that challenge our traditional notion of *community*;
b) some VEs and VR applications allow users to construct new and alternate (personal) *identities*;
c) AI-related developments invite us to reassess our *sense of self* and, ultimately, question *what it means to be human* in a world that we share with non-human "intelligent" agents and entities.

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Learning Objectives

Upon completion of this lesson the students should be able to:

- Discuss how online communities such as Facebook and X have caused us to rethink traditional concepts of "community"
- Describe how interactions occurring on social networking and online forum sites have affected our conventional notions of friendship
- Describe and discuss concerns about social and ethical behaviors arising from interactions in massively multiplayer online role-playing games (MMORPGs)
- Discuss the implications of role-playing in online environments for our understanding of personal identity
- Explain how developments in Artificial Intelligence (AI) have affected our sense of self and our conception of what it means to be a human being
- Discuss the issue of whether certain types of AI entities warrant moral consideration

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PI: Online Communities

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Module I Objectives

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1. Communities in Cyberspace

- *Webster's New World Dictionary of the American Language* defines a community as "people living in the same district, city, etc., under the same laws."
- Note that this (traditional) definition of community – with its emphasis on people living in the "same district" or "same city," etc. – stresses the importance of geography and physical space.
- But our traditional conception of community has evolved significantly because of our interactions in online forums and social networking sites (SNSs) such as Facebook.

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Forming Online Communities

- In the past, many people tended to identify themselves as members of a community by way of categories such as their national heritage, religious affiliation, etc.
- Today, *common interests*, as opposed to traditional categories, often bring people together to form *online communities*.

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Are you a part of an Online Community

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Satisfaction for Those in Online vs. Traditional Communities

- Parsell (2008) cites a survey showing that 43% of members of online communities claimed to feel "as strong" about their online communities as their traditional or "real world" communities.

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Blogs as Online Communities

- political blogs,
- corporate blogs,
- health blogs,
- literary blogs,
- travel blogs.

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Pros and Cons of Online Communities

Positive Effects include

- meet new friends and future romantic partners;
- form medical support groups by joining Internet-based social networking groups;
- join discussion forums to disseminate material to like-minded colleagues;
- communicate with people they might not otherwise communicate with by physical mail or telephone.

Negative effects because they threaten traditional community life and minimize face-to-face communications.

- facilitated *social polarization*,
- threatened our traditional notion of *friendships*,
- facilitated *deception*.

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Social Polarization (Parsell)

- People tend to be attracted to others with like opinions.

1. Being exposed to like opinions tends to increase our own prejudices.
2. This polarizing of attitudes can occur on socially significant issues…
3. Where the possibility of narrowing focus on socially significant issues is available, increased community fracture is likely.

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Online Communities and the Impact for *Friendships*

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Friendships in *Online-Only* Communities

- What implications do online-only communications between individuals have for our traditional understanding of *friendship*.
- Is it possible for people who interact only in virtual (or purely online) contexts to be "real friends"?

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Friendships in Online-Only Communities (Continued)

- Cocking and Matthews (2000) argue that the kinds of close friendships we enjoy in physical space are not possible in pure virtual environments (i.e., in contexts that are solely computer-mediated).
- They note that online-only friendship occurs in "a context of communication dominated by *voluntary* self disclosure, enabling and disposing me to construct a highly chosen and controlled self-presentation."
- As a result, Cocking and Matthews believe that we "miss the kind of interaction between friends that seems a striking and commonplace feature of a close friendship."

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Online Friendships (Continued)

- Briggle (2009) disagrees with Cocking and Matthews.
- Briggle considers factors such as "sincerity" and "distance" in making his case for why friendships in purely virtual contexts can be "initiated and flourish."
- He points out that communications among friends in off-line contexts, which are based largely on "oral exchanges," are not always candid or *sincere*.
- Briggle also notes that the *distance* involved in typical computer-mediated communications can give friends the courage to be more candid with one another than in typical face-to-face interactions.

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Online Communities and *Deception*

- Online communities can also reveal a "darker side" of the Internet because people can, under the shield of anonymity, engage in behavior that would not be tolerated in most physical communities.
- For example, individuals can use aliases and screen names when they interact in online forums, which makes it easier to *deceive* about who actually is communicating with them.

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Ethical Concepts and Theories

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Ch11: Online Communities, Virtual Reality, And Artificial Intelligence
P2: Virtual Worlds & RPGs

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Module Objectives

Upon completion of this lesson the students should be able to:

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Defining "Virtual" in Virtual Environments

1) contrasted with "real," as in cases where virtual objects are differentiated from "real objects";
2) contrasted with "actual" – e.g., where a person claims that she is "virtually finished" her project;
3) used to express a feeling that one has "as if" he or she were physically present – i.e., the feeling of presence that one can have in a telephone conversation or an online messaging exchange.

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Virtual Environments (Continued)

- As already noted, VEs exist in *virtual space*.
- What is meant by "virtual space"?
- Virtual space can be viewed as a range of *computer-generated environments* that could not exist without computers and cybertechnology.
- These environments, in turn, include online (or virtual) communities, as well as (three-dimensional) virtual reality (VR) applications.

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VEs and Virtual Worlds

- Not only do VEs subsume virtual (or online) communities, they also subsume (what some authors call) *virtual worlds*.
- What are virtual worlds, and how are they similar to and different from VEs.
- Søraker and Brey describe virtual worlds as a type of VE in which users typically:

a) are "represented by avatars";
b) have the "illusion of perceiving a three dimensional world consisting of virtual objects."

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Defining Virtual Reality and VR Applications

- What, exactly, is meant by the expressions "virtual reality" and "VR application"?
- Brey (1999) defines virtual reality (VR), as well as a VR application, as

"*a three-dimensional interactive computer-generated environment that incorporates a first-person perspective*." [Italics Added]

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Virtual Reality vs. *Augmented Reality*

- So, VR applications typically provide the user with a "simulated world," thereby replacing the real world.
- AR technology, on the contrary, enhances (or *augments*) a user's view of the real world.

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Ethical Aspects of VR Applications

- Brey also notes that VR users are not spectators, but rather are more like actors, i.e., as are the board game players who also act out roles in certain board games.
- But Brey also points out that VR applications, unlike board games, simulate the world in a way that gives it a much greater appearance of reality.
- In VR applications, the player has a first-person perspective of what it is like to perform certain acts and roles, including some that are criminal or immoral, or both.
- We next examine some ethical aspects of one type of VR application: online video games.

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Ethical Aspects of Online Video Games

- We should note that some analysts differentiate single-player video games from multi-player online games, which are commonly referred to as MMORPGs (Massively Multiplayer Online Role-Playing Games).
- MMORPGs include games such as *Second Life* and *World of Warcraft*.
- *Second Life* (designed by Linden Lab), which includes members called "Residents," (has 1million regular users as of 2014, and 13 million registered accounts as of 2008).
- *World of Warcraft (WOW)*, one of the most popular MMORPGs, has 7 million subscribers as of Aug. 2014.

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Violent and Sexually-Offensive Acts in MMORPGs

- Some critics claim that Second Life facilitates child pornography because virtual characters who are adults in real life can have sex with virtual characters who are children in that MMORPG (Singer, 2007).
- Wonderly (2008) suggests that some forms of violence permitted in online games be "more morally problematic" than pornography and other kinds of sexually offensive behavior in virtual environments.

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Assessing "Harm" Resulting from Acts in VEs and VR Applications

- Is it wrong to perform acts in VEs and VR applications that would be considered immoral in "real life"?
- Can these acts cause *real harm*?
- We might think that since no one can be physically harmed in a VE, any harm caused in the virtual realm is not "real harm" but only *virtual harm*.

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Arguments for Evaluating Harm in Virtual Environments/VR Applications

- Brey describes two kinds of arguments that can help us to evaluate harm in virtual environments:

1. The *argument from moral development*, which reasons that the way we treat virtual characters can affect the way we treat real-life people.
2. The *argument from psychological harm*, which reasons that the way we interact with virtual characters can cause psychological harm to people in real-life situations who have suffered harm (e.g., as in the case of Lambda-Moo and real-life rape victims).

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Virtual Economies and "Gold Farming"

- Kimppa and Bisset (2008) define gold farming as "playing an online computer game for the purpose of gaining items of value within the internal economy of the game and selling these to other players for real money."
- These "virtual" items can include "desirable items" as well as in-game money (where the rules defining the game's internal economy permit this).
- The items can also include "highly developed" game characters.

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Virtual Economies and "Gold Farming" (Continued)

- In some cases, gold-farming practices have become advantageous to the "gold farmers" themselves, who were able to earn more money obtaining and selling virtual gold than they could than in traditional agricultural work.
- But gold-farming practices have also led to reported cases of Chinese sweatshops, where laborers can work "day and night in conditions of practical slavery" to acquire the virtual gold and virtual resources (Brey 2013).

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Ethical Concepts and Theories

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Ch11: Online Communities, Virtual Reality, And Artificial Intelligence
P3: AI, Humans, & Cyborgs

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Learning Objectives

Upon completion of this lesson the students should be able to:

- Explain how developments in Artificial Intelligence (AI) have affected our sense of self and our conception of what it means to be a human being
- Discuss the issue of whether certain types of AI entities warrant moral consideration

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3. Artificial Intelligence (AI) and its Implications for What it Means to be Human

We can ask: Which criteria distinguish humans from other kinds of creatures and entities?

- Is *rationality* a plausible criterion?
- The view that only humans are rational has recently been challenged on two separate fronts:

1) research in animal behavior/intelligence suggests that many primates, dolphins, and whales are capable of demonstrating skills we typically count as rational;
2) developments in artificial intelligence (AI) have shown that certain forms of "rational activity" can also be attributed to computers.

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What is AI?

- John Sullins (2015) defines AI as "the science and technology that seeks to create intelligent computational systems."
- He notes that AI research has aimed at building computer systems that can duplicate, or at least simulate, the kind of intelligent behavior found in humans.

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AI Background/Developments (Continued)

- John Weckert (2001) articulates this kind of (sociological) concern when he asks:

> Can we, and do we want to, live with artificial *intelligences*? We can happily live with fish that swim better than we do, with dogs that hear better, hawks that see and fly better, and so on, but things that can reason better seem to be in a different and altogether more worrying category...What would such [developments mean for] our view of what it is to be human?

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Can We build (Genuinely) *Intelligent* Machines? An Ongoing Debate

- We can ask whether it is possible, even in principle, to build "machines," i.e. either software programs or (physical) artificial entities, that are "genuinely intelligent" and whose intelligence could rival and possibly exceed that of humans.

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The *Turing Test* as a "Thought Experiment" for Machine Intelligence

- In 1950 Alan Turing predicted that by the year 2000, a computing machine would be able to pass a test, which has come to be called the *Turing Test*, demonstrating machine intelligence.
- Turing envisioned a scenario in which a person engaged in a conversation with a computer (located in a room that was not visible to the human) was unable to tell – via a series of exchanges – whether he or she was conversing with another human or with a machine.
- He believed that if the person could not be sure that this entity was a human or a computer, then we would have to attribute some degree of intelligence to the computer.

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The *Turing Test- GPT, LlaMa*

- A March 2025 study from UC San Diego researchers evaluated several AI systems in controlled, three-party Turing tests.
- GPT-4.5, when prompted to adopt a human-like persona, was judged to be human 73% of the time — actually more often than the real human participants.
- LLaMA-3.1 was judged human 56% of the time
- The researchers called it "the first empirical evidence that any artificial system passes a standard three-party Turing test."

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The *Turing Test* and Searle's "Chinese Room" Argument

- Was Watson merely acting in a manner similar to the individual in John Searle's (now) classic "Chinese Room" thought experiment?
- In that scenario, a human who is a native English speaker but who understands nothing about the Chinese language is able to perform tasks that require manipulating Chinese symbols to produce correct answers to questions posed in Chinese.

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The "Chinese Room" Argument (Continued)

- The person in the "Chinese room," who is not seen by anyone outside that room, receives questions from someone who passes them to him through an opening or slot.
- That person then consults a series of instructions and rules located inside the room, all of which are written in English.
- The person is able to substitute the incoming Chinese symbols for other Chinese symbols in such a way as to produce correct answers to the questions asked.

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The Chinese Room Argument (Continued)

- Once the person inside the room has completed the task, he passes the answers through the slot to a person waiting outside.
- That person might assume that the individual who returned the correct answers understood Chinese.
- But Searle (1980) argues that it is possible that the person (inside the room) understood nothing about the semantic meaning of the questions he received and the answers he returned.

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The Chinese Room Argument (Continued)

- Searle points out that the person in the room could merely have been following a set of syntactic rules (written in English) that were designed to manipulate symbols that happened to be in Chinese.
- In fact, it is possible that the English-speaking person (inside the room) may not even know that the symbols involved are elements of the Chinese language.

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Cyborgs and Human-Machine Relationships

a) Are humans becoming more computer-like?

b) Are computers becoming more human-like?

- We look at (a) from the perspective of AI-induced bionic chips.

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Cyborgs and (AI-induced) Bionic Chip Implants: Are Humans Becoming More Computer-Like?

- Future chip plants made possible by AI could be designed to make a normal person "super human."
- Weckert (2001) notes that "conventional" implants designed to "correct" deficiencies have been around and used for some time.
- The purpose of conventional, or "therapeutic" implants, has been to assist patients in their goal of achieving "normal" states of vision, hearing, heartbeat, etc.

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AI and Controversies Involving Bionic Chip Implants (Continued)

- Moor (2005) notes that while therapeutic implants will likely be accepted, "enhancement implants" will likely be controversial.

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AI and Chip Implants (Continued)

- In other words, Moor believes that:
- therapeutic chip implants such as pacemakers, defibulators, and bionic eyes that maintain and restore natural bodily functions will most likely be accepted;
- enhancement implants, such as giving patients added arms or infrared vision, will most likely be prohibited.

AI and Chip Implants (Continued)

Additionally, Moor believes that such a policy would likely endorse the use of chip implants that:

- reduced dyslexia;
- assist memory of Alzheimer patients.

But Moor also believes that this kind of policy would not endorse the implanting of either a:

- "deep blue" chip for superior chess play;
- miniature digital camera that would record and playback what a person had just seen.

Are We (Becoming) Cyborgs?

- Some now worry that with bionic parts, humans and machines could soon begin to merge into *cyborgs*.
- Ray Kurzweil (2000) suggests that the distinction between machines and humans may no longer be useful.
- James Moor (2005) believes the question we must continually reevaluate is not whether we should become cyborgs, but rather: *What sort of cyborgs should we become?*

Are Computers Becoming More Human-Like (Continued)?

- Sophisticated robots of the future may not only look more human-like, but may also exhibit sentient characteristics.
- These robots, like humans and animals, could (arguably) have some *sentience* (and thus be capable of simulating the experiences of sensation, feeling, and emotion).
- Robots and other kinds of AI entities of the near future may also exhibit, or appear to exhibit, consciousness.

Do (At Least Some) AI Entities Deserve Moral Consideration

- Three questions worth considering from Scenario 11-4 are:

1) Should the artificial boy (or any artificial creatures like "him") ever have been developed in the first place?
2) Is it morally permissible for the "boy's" adopted parents to discard "him" as they would some other kind of "computer resource"?
3) Does this "boy" deserve (at least some) *moral consideration* (especially from the human parents who adopted "him")?

Expanding Our Sphere of Moral Consideration (Continued)

- Prior to the 20th century, we generally assumed (for whatever reasons) that *only* human beings deserved ethical consideration.
- All other entities—animals, trees, natural objects, etc.—were assumed to be mere *resources* for humans to use (and abuse) as they saw fit.

Expanding Our Sphere of Moral Consideration (Continued)

- By the mid-twentieth century, the assumption that moral consideration should be granted only to humans was challenged by both:
- animal-rights activists/groups,
- environmentalists.

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Expanding Our Sphere of Moral Consideration (Continued)

- Animal-rights advocates point out that animals, like humans, are capable of feeling pain and suffering (i.e., they are *sentient*).
- Many of these advocates also argued that because animals can suffer, we should grant them moral consideration.

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Expanding Our Sphere of Moral Consideration (Continued)

- Influenced by the work of Hans Jonas (1984), some philosophers and many environmentalists have argued that we should also extend our realm of moral consideration to include:
- trees,
- plants,
- the entire ecosystem.

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Expanding Our Sphere of Moral Consideration (Continued)

- Will some AI entities eventually qualify as *moral agents*?
- Even if AIs are unable to qualify as full-blown moral agents (as most adult humans do), they may nevertheless meet the threshold of what Floridi calls "moral patients."
- In Floridi's scheme, moral patients are "receivers of moral action" (while moral agents are the "sources of moral act- ion" – i.e., capable of causing moral harm or moral good).
- Like moral agents, moral patients enjoy moral consideration and thus have at least some moral standing.
- Unlike moral agents, however, moral patients cannot be held morally responsible for their actions.

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